6	**(a)**	$E = 0$ *or* $E_A = (–)E_B$ (at $x = 11\,\text{cm}$)	B1	
		$Q_A / x^2 = Q_B / (20 – x)^2 = 11^2 / 9^2$	C1	
		Q_A / Q_B or ratio = 1.5	A1	[3]
		or		
		$E \propto Q$ because r same *or* $E = Q / 4\pi\varepsilon_0 r^2$ and r same	(B1)	
		$Q_A / Q_B = 48 / 32$	(C1)	
		Q_A / Q_B or ratio = 1.5	(A1)	

(b) **(i)**	for max. speed, $\Delta V = (0.76 - 0.18)$ V or $\Delta V = 0.58$ V	C1	
	$q\Delta V = ½mv^2$		
	$2 \times (1.60 \times 10^{-19}) \times 0.58 = ½ \times 4 \times 1.66 \times 10^{-27} \times v^2$	C1	
	$v^2 = 5.59 \times 10^7$		
	$v = 7.5 \times 10^3\,\mathrm{m\,s^{-1}}$	A1	[3]
(ii)	$\Delta V = 0.22$ V	C1	
	$2 \times (1.60 \times 10^{-19}) \times 0.22 = ½ \times 4 \times 1.66 \times 10^{-27} \times v^2$		
	$v^2 = 2.12 \times 10^7$		
	$v = 4.6 \times 10^3\,\mathrm{m\,s^{-1}}$	A1	[2]